

The More-than-Human Talking Stick: Design Principles for Multispecies Entanglement and Agency

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Figure 1: Left to right: A: fieldwork in the community garden. B: interviewee showing a specific plant. C: the designed artefact *Mth talking stick*. D: inside view of the artefact.

Abstract

This paper explores how multispecies agency and relations can be brought to the foreground in a situated design process through utilising design principles. Based on an analysis of fieldwork at a permaculture garden community, our three design principles – consideration, connectedness, dynamics – proved to be useful both as a way to guide the design process towards a designed artefact and to inform continued awareness of multispecies agency and entanglements. The resulting design, The Mth Talking Stick, is an attempt to create an artefact that intervenes with a more-than-human voice in mostly-human conversations. The paper contributes situated insights into how design principles can strengthen human awareness of multispecies entanglements as a response to calls for stronger empirical foundations in the more-than-human design field.

CCS Concepts

• **Human-centered computing** → **Contextual design**.

Keywords

HCI, More-than-human, Design

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1 Introduction

Although posthuman theory has made clear the need to understand and recognize distributed agency and relationality [4, 19], design research still struggles to provide practical approaches to make these ideas actionable: While we are witnessing a shift away from human-centered thinking, it is stated that the growing field of more-than-human (Mth) design is "still of a theoretical or speculative nature" [6] in need of a stronger empirical foundation and a more



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cohesive understanding of the field and its practices [6, 14]. Without situated empirical exploration, MtH design risks remaining a theoretical aspiration of decentering the human. In this paper, based on a study with a permaculture garden community, we explore how *design principles* can guide design that strengthen human awareness of multispecies entanglement.

Anchored in posthumanism, contemporary design approaches are increasingly interested in MtH epistemologies, which heightens focus on human and nonhuman entanglements as a critique of human-centered thinking [6, 34]. Herein, it can be understood that all actors and actor types have agency – distributed, entangled, and (in reference to Barad [4]) intra/related. [15, 16, 28]. Some contributions have explored how MtH thinking shaped specific design practices and projects: In their research on smart cities, Heitlinger et al. [20] explores how sensitivity towards MtH affect rights-based approaches. Rolighed et al.'s Plant Radio explores surfacing nonhuman activity to transform how humans understand and are sensitive to plants [33]. Some researchers express how design practices are key to understanding MtH relations, for instance Light [23] explores how being “in dialogue” with the MtH can shape and build sensitivity, which is also discussed by [17]. Following this approach of sensitivity, Rodgers et al. [32] shows how practices of attention and awareness towards MtH teach designers new ways of engaging with their nonhuman relations. By working “in the dirt”, grounding action in the material and ecological circumstances of a place, Jack [21] makes clear how relational awareness and engagement shape method and outcome. Similarly and lastly, with respect to participatory approaches, the contributions in [26] show how including nonhuman actors in making changes the frame of understanding participation.

Garden communities are examples of complex entanglements between humans, nonhumans, technologies, and environments. Some garden communities seek to query anthropocentric and human-centered thinking and are emerging as local grassroots’ responses to the accelerating biodiversity and climate crises, combining social practices and ecological awareness, exploring ways of building social and ecological resilience [21, 25]. In this study, inspired by [30, 36, 39, 40], we engage with a garden community as part of a larger research project where we explore entanglements as key in designing artefacts that shape human sensitivity towards nonhumans. Understanding MtH communities – and in this particular case a permaculture garden community – as dynamic, relational assemblages allow us to see how every involved entity (i.e. humans, nonhumans, technologies, and environments) is co-constituted through ongoing entanglements. Volunteer communities like these are characterized by distributed agency where meaning and action are co-produced across all actors, and the role of technology is often to mediate shared activities in dynamic artifact ecologies [8, 9, 22]. This perspective frames communities not as static entities but as evolving collectives of interdependent actors whose relations are constitutive of both social and material worlds.

In the following, we report on the process of moving from fieldwork in the garden community to a design, and illustrate how phrasing design principles strengthened our designerly sensitivity towards multispecies entanglement. We discuss future implications of formulating design principles based on situated knowledge in the MtH design process.

2 Case study: More-than-Human Communities

In order to explore design possibilities in the context of a more-than-human community, we initiated a design process where notions of MtH inclusivity, awareness, and agency were already foregrounded as theoretical aspirations for the MtH designer [10, 35]. However, when we moved from theoretical work to empirical work, these aspirations were often hindering our design decisions, as they were difficult to live up to. For instance, incorporating non-human agency in a design made by humans somehow requires bypassing one’s own human positionality, which is easier to say than do. This caused moments of frustration and the feeling of being stuck in a loop of paradoxes, while also shaping a dynamic and co-constitutive process where negotiations between the needs of humans as well as MtH became foundational for the design process.

Using multispecies ethnographic fieldwork, which demands a heightened sense of noticing of the nonhumans’ presence in the field [7, 27], we toured the permaculture garden community with one of its members, see Figure 1A-B. The garden community is located in Aarhus, Denmark and was founded in 2023 with approximately 20 members, where some members tend to their own raised beds and others are more engaged in the community’s permaculture areas. Our interviewee had been a member since the beginning, and our tour lasted 1,5 hour and was carried out in mid October. After the visit, we analyzed the fieldwork data by defining general themes, following the first four phases of reflexive thematic analysis: familiarization, coding, generating initial themes, and developing and reviewing themes [11, 12]. The idea was to make the fieldwork inform the design process, casting the data as inspiration and as a tool to transitioning from analysis to design. The analysis resulted in three clustered themes.

Theme 1: inclusion, availability, consideration This theme emerged through our interviewee’s stories about community negotiations on the member guidelines. For instance, a suggestion that all soil, compost, and seeds must be organic was rejected, as organic products are expensive and would thus exclude members with fewer resources. Furthermore, many members get plants and seeds by trading with others, making it impossible to trace lineage and herbicidal provenance. Another suggestion was to introduce commitment rules, which was rejected as community members are diverse in age, abilities, and stages of life, and thus have differing levels of commitment and energy to put into the garden. Members’ motivation for being part of the community also varies. “*you meet people with similar interests even though we are very different. Some members are retired and others still work. [...] Looking at the plants, they are very big here [at the community area]. They could never be this big in my tiny garden. But [a member] contributes with different perspectives, for example inspiration and knowledge.*” The garden community is an opportunity for human relations, but also for exploring nonhuman relations that the member’s own garden could not facilitate.

Theme 2: network, co-dependency, connectedness Dependencies were also evident in the participant’s emphasis on how they seek to continuously work with the MtH: “*We observe and try to grow with nature and not against it. We take care of the soil as a place where animals and organisms live and try to keep it moist and alive.*” Our interviewee also emphasized how the garden space was a place to

socialize and network with others, while also reflecting on their co-dependency with the plants: "*Because we have planted plants in the community area, we are dependent on the community continuing.*" This highlights how the community operates as a living assemblage where practices of care and awareness shape both human and nonhuman development.

Theme 3: chaos, spontaneity, dynamic As we entered the garden, we immediately encountered practices where human-nonhuman relations were apparent. The Hügelskultur beds, ponds, insect hotels, and edible perennials are just some of the features that indicate multispecies coexistence. These elements embodied the entanglement and relationality that permeate our theoretical framing, with distributed agencies across human and nonhuman actors [3, 15, 16] being shown through the way things generatively emerge and disappear, take-over and die out. Themes of spontaneity, chaos, and dynamics emerged in multiple descriptions of how the community works with its environment and its nonhumans: "*A lot of things just happen spontaneously*" and "*We [the community] are also a little chaotic, so you can also just do what you want.*" The community works with what it has: moving plants if they grow larger than expected; building a small pond with leftover materials just because a member happened to have these, etc.

In the design process, before formalizing these themes further into forms and functionality, we experienced a need to combine and ground the fieldwork to the project's posthuman aspirations, which resulted in formulating three initial design principles to keep in mind for the rest of the design process:

- (1) The design must be *considerate* of different needs. Based on how permaculture members differ in levels of commitment, resources, energy, and purposes, and how the community is also made up of many other-than-human species, the design cannot demand too much effort, or exclude some from the community.
- (2) The design must reflect *connectedness*. It should express multispecies entanglements and represent human interconnectedness with the nonhumans, as this co-dependency and relatedness was very present in our tour.
- (3) The design must be *dynamic*. As the garden operates with living matter that undergoes constant changes, the design should also reflect this by showing change over time – be "living" rather than static.

Attending to the three principles created awareness of and attunement to the nonhumans while also considering human users. We further extended the principles with metaphors that correspond or relate to the themes. This led to the design of the *MtH Talking Stick*, an interactive artefact inspired by both the authority symbolised in a royal sceptre and the permission to speak given in the indigenous formal practice of a speaker's staff or a talking stick.

2.1 The More-than-Human Talking Stick

The More-than-Human Talking stick is an artefact asking for human attention to remind them of both human influence over nonhumans and the agency of nonhumans, see Figure 1C-D. Using an omnidirectional microphone and an offline speech-to-text recognition toolkit [1], the artefact uses wake word detection of the word "we"

to trigger a solenoid. When the wake word is detected, a Raspberry Pi converts it into an analog output, triggering the solenoid to hit into a wooden percussion box, thus imitating the thumping sound of knocking a wooden talking stick. The artefact is cylindrical (30x10cm), has a 3D printed base and top, and a laser cut lattice hinge wood body [31]. The technology and functionality are hidden to allow the artefact to fit into multiple situations and settings where decisions affecting nonhumans are made. This could be e.g. community garden planning meetings, teaching, workshops or summits, where MtH entanglements are focal points and in need of continuous attention.

The talking stick metaphor – an object symbolising the power and right to be recognised and speak in a gathering – has shaped our prototype's cylindrical form, natural materials, and most importantly its functionality of "responding" with a wooden knock to create a brief interruption in a conversation, thus catching people's attention. The effect is similar to a subtle "ahem" or fake cough to attract attention or imply disapproval.

This redistribution of attention upon hearing the word "we" is thought to create room for reflection without speaking directly on behalf of specific nonhumans. The sound of wood was similarly chosen to not represent any specific nonhuman sound, while still having ties to organic materiality and being a physical, situated action. With speech as input type, the artefact also signifies how human language has immense power [3]. The instant response of the wake word detection positions the artefact as a situational reminder to extend the meaning of "we" to cover more-than-human communities as well, not just "we humans", since human decisions and actions have direct impact on the environment. The wake word "we" signals deep social meanings, collective identity, and responsibility often seen in volunteer communities [22] and it symbolises entanglement, relations, and connected responsibility without explicitly representing any specific human or nonhuman. On triggering this wake word, the artefact seeks to push the boundaries of the word "we" and train our understanding of sociality to also include the other-than-humans, a concept that we meet in more-than-human theories such as Tsing [37].

2.2 Applying design principles in practice

In terms of *dynamic*, the only feedback in our current prototype is the sound of the solenoid moving inside the artefact. Its ambiguous form does not disclose where the sound comes from or that it correlates to the wake word, resulting in an artefact that feels like it can "act on its own" and is unpredictable and "alive".

In terms of *consideration*, our design has a low entry point and requires low maintenance; users can simply plug it into power and let it be. It is stand-alone, allowing it to be moved around, fitting into various settings and existing artifact ecologies [8, 9]. However, the design can be exclusionary as it is based on an English speech recognition model, making it difficult for non-native speakers and impossible for nonhumans to be recognised. The speech input can also be affected by ambient noise, which may lower the effectiveness of the design. Another aspect tying to consideration is privacy: as the microphone captures audio this might raise concerns as to whether it is a piece of surveillance technology, although it is offline and does not store any data.

The artefact expresses *connectedness* in different ways. By reminding the human user that "we" encompasses more than just human entities, MtH agency and presence is brought to attention and "inserted" into mostly human gatherings, thereby demonstrating that we – humans and nonhumans – are all connected in interacting and inter-dependent ways, as suggested by [4, 18]. As a physical object, the presence of the artefact in a room creates a symbolic connection to the outside garden environment as it signifies and speaks on behalf of the more-than-human entities. While we recognise that nature is all around and in us, and that a distinction between inside and outside makes little sense in a MtH context [13], we experiment with unlearning this dualistic worldview by bringing attention to these two positions through the design artefact: a human talks, something instantly makes itself known. Lastly, as the design does not represent a specific nonhuman (e.g. by taking the shape of a bird or a snail), it could be argued that it creates a connection to everything nonhuman.

The artefact only partly lives up to the principles, but does allow for them to be open-ended and come into play from different perspectives in different ways, e.g. how *consideration* was originally derived from the garden community's different commitment levels, but can also be about considerations for privacy; or how *dynamic* is mostly referring to change for our visual senses, but can also be manifested through sound and other kinds of liveliness. Despite these shortcomings, our case suggests that although specific principles do not automatically inform expected outcomes they can act as tools for grounding the designer in being aware and reflexive in MtH research. We find that formulating design principles based on fieldwork is one way to answer the call for a stronger empirical foundation and a deeper understanding of the field.

3 Discussion and future work

Our experience with needing, formulating, and applying guiding design principles presents as an interesting dilemma in MtH design research. In trying to design in, to, and with a community that extends beyond the merely human and includes the nonhuman, we as designers are met with our own humanness and challenged by it. Also, in wanting to speak on behalf of the nonhumans, we also have to be considerate not to silence the humans. Furthermore, when considering the agency of a designed technology implemented in a human context, we also must balance the interventional power of the artefact with the power of the human to simply turn it off. The aspiration of bringing distributed agency and inclusiveness for all to the surface seldom marries perfectly with (human) usability and context practicalities. For example, we intentionally chose to let the artefact have an ambiguous form while still being attention grabbing, designing the look and sound to subtly interrupt but not entirely overtake and halt existing conversation. A more intrusive artefact might have played into giving more authority to nonhumans, but might also have disrupted human use and acceptance of its presence.

In a design process, these challenges and paradoxes can be hindering and difficult to work around, and in our case the design principles helped guide and ground us in this complex design landscape. We do not think of these three particular principles - consideration, connection and dynamics - as universally applicable, although they

might also work for others. We call for other designers to create their own set of principles in a design process involving nonhumans as a method to uphold attentiveness towards the nonhuman in an otherwise very human design process. Naming it "design principles" might seem too constrained, rigid and universal, but synonyms such as design rules, values, guidelines, ideals, and aspirations seemed not to fit.

Having organizing principles in a design process is not new to the field of interaction design and HCI. For instance, in a more formalized sense they have been referred to as "visions": *"When a designer is confronted with a design situation, a vision emerges. [...] It should be thought of as a first organizing principle that helps the designer to structure the initial attempts to respond to the situation at hand."* [24, p.17-18]. However, visions are not necessarily related to a designer's ethics, morals, and ideologies, which working in the MtH realm demands. In our case, ethics and care was implicitly or peripherally addressed, but as Barad's ethico-onto-epistemology perspective suggests, matterings involving the human and nonhuman cannot avoid ethics, as *"delicate tissue of ethicality runs through the marrow of being"* [5, p. 396]. Hence there is a need to expand on this concept of a guiding vision that also considers the ethical aspects of being a MtH design researcher.

The design principles have shaped specific design choices, while also shedding light on where the design only partly fulfills them. The principles were also vague enough to allow for new interpretations and perspectives to emerge after the design process. They have been a way to ground ourselves as designers in our aspirations, while also being an analytical tool to further review both our design and the principles themselves. In other words, the principles have helped us to *stay* with the trouble in two ways: 1. it helps us stay – in contrast to drifting away or escaping – from the paradoxes, and pushes us to work through them, and 2. staying – as in keep being conscious – with respect to how the design does not fully live up to the principles and is definitely also an on-going open-ended negotiation with things and MtH worlds as Wakkary [38] suggests.

4 Conclusion

This paper responds to calls for a stronger empirical foundation and a more cohesive understanding of the MtH design field. We do this by contributing situated insights formulated as design principles that can place MtH awareness centrally in design processes. Formulating design principles grounded in field studies proved valuable for our process. Similar to other design methods such as e.g. non-human personas [2, 29], the principles did not guarantee specific outcomes but kept MtH awareness central throughout the design process. The principles acted as a guide for navigating complex layers of agency and they created situations that caused both hesitation and reflection. From this, we see how our principally informed design process holds value over our artefact outcome. Although there are several limitations in the case study presented here, it contributes with an illustration of how having design principles to guide a MtH design approach can act as tools for more-than-human awareness both during and after the making of the actual design.

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